



SR 816.111

Ergänzung 2.3 zu Anhang 5 der Verordnung des EDI vom 22. März 2017 über das elektronische Patientendossier

Nationale Integrationsprofile nach Artikel 5 Absatz 1 Buchstabe c EPDV-EDI

Community Portal Index (CH:CPI)

Ergänzung 2.3 zu Anhang 5 EPDV-EDI : Nationale Anpassungen

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1 Introduction

Dieses Dokument wurde erstellt, um die schweizerischen Regelungen der Verordnung über das elektronische Patientendossier (EPDV, SR 816.11) zu erfüllen. Die EPDV und die EPDV-EDI (SR 816.111) werden in der Amtlichen Sammlung veröffentlicht (in Deutsch, Französisch und Italienisch)¹.

Das elektronische Patientendossier (EPD) basiert auf einem System mit mehreren (Stamm-)Gemeinschaften, in welchem die Patientin oder der Patient nicht nur die Zustimmung zur Erstellung und Verwendung ihres oder seines Dossiers gibt, sondern auch explizit Zugangsregeln durch ein Zugangsportal für Patientinnen und Patienten festlegt.

Die Architektur dieses Systems bildet daher einen Schweizer EPD-Vertrauensraum und alle teilnehmenden (Stamm-)Gemeinschaften müssen Zugang zu einem vertrauenswürdigen Index eines zertifizierten Partners haben.

Das vorliegende Dokument spezifiziert das entsprechende nationale Integrationsprofil, genannt "Community Portal Index (CH:CPI)". Es enthält derzeit nur Informationen über (Stamm-) Gemeinschaften, da Portale derzeit nicht Teil der Schweizer EPD sind.

This document fulfils the Swiss regulations of the Ordinance on the Electronic Patient Record (EPRO, SR 816.11). The EPRO and the EPRO-FDHA (SR 816.111) are published in Official Compilation of Federal Legislation (available in German, French and Italian)¹.

The Swiss Electronic Patient Record (EPR) depends on a multi (reference) community-based system where the patient not only consents to the creation and use of the record but does so by explicitly defining access rules through a patient portal.

The architecture of this system builds therefore a Swiss EPR circle of trust and all participating communities need to have access to a trustworthy index of certified partners.

The present document specifies the corresponding national integration profile, called "Community Portal Index (CH:CPI)". It currently contains only information about communities, as portals are not currently part of the Swiss EPR.

¹ German: <https://www.admin.ch/opc/de/classified-compilation/20111795/index.html>;
French: <https://www.admin.ch/opc/fr/classified-compilation/20111795/index.html>;
Italian: <https://www.admin.ch/opc/it/classified-compilation/20111795/index.html>.

2 Volume 1 – Integration Profile

2.1 Overview

The **Community Portal Index (CH:CPI)** is an index containing all information about certified (reference) communities and their endpoints according to the Federal Act on the Electronic Patient Record (EPRA). The CH:CPI-Service is one of the central services provided by the Federal Office of Public Health (FOPH) and the FOPH is also responsible for the content of the CH:CPI.

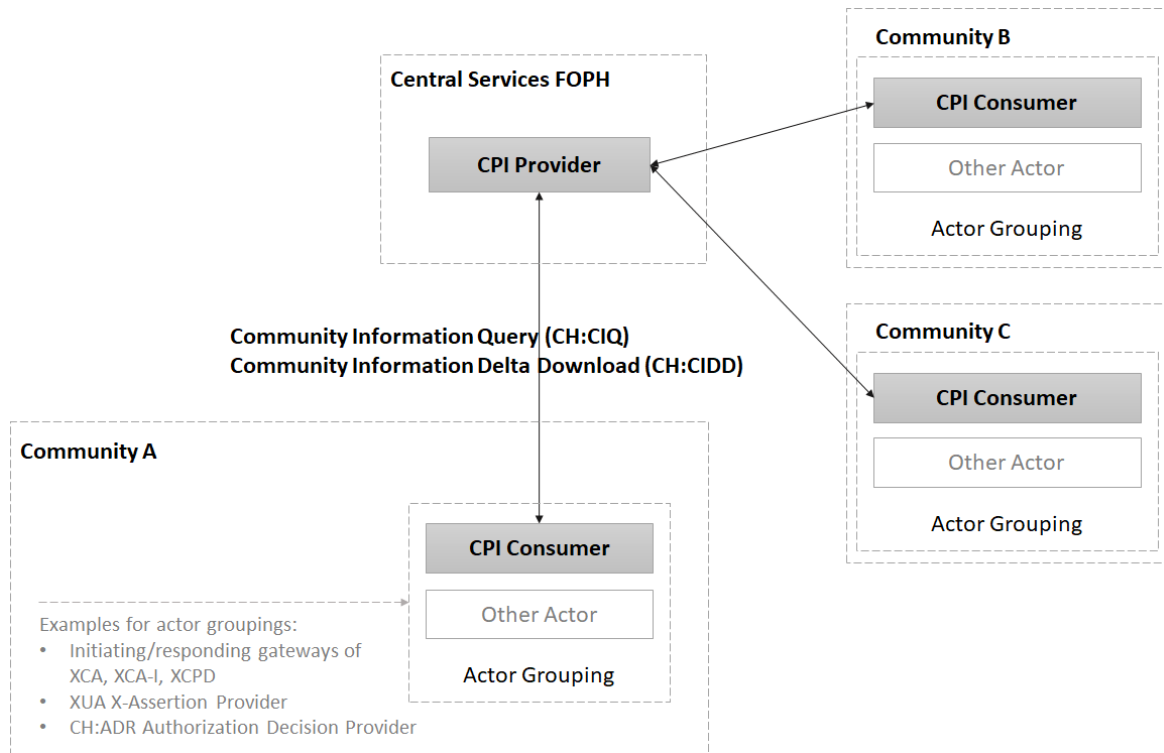


Figure 1: CH:CPI Actors within the Swiss EPR circle of trust

2.2 Community Portal Index (CH:CPI)

This profile defines functionalities for the Swiss EPR circle of trust in order to provide a trustworthy list of certified communities and reference communities within the Swiss EPR circle of trust. This list also contains information about relevant endpoints of the certified (reference) communities.

In other words, the content of the CH:CPI is the binding endpoint configuration list and therefore allows the trustworthy cross-community addressing among the certified Swiss EPR participants.

2.2.1 Motivation

The Swiss EPR is based on multiple integration profiles (IHE Profiles and national profiles) that can be split in community internal profiles and cross-community profiles. Using the CH:CPI, all (reference) communities as well as the central services get trustworthy connected. The productive CH:CPI is for security reasons only accessible by certified communities and the FOPH administrators.

2.2.2 Objectives and Constraints

The objective of the CH:CPI Profile is the definition of a mechanism to request the endpoint configurations within the Swiss EPR circle of trust. The defined mechanism follows the IHE HPD profile², as both profiles follow the same communication requirements but different content.

² See Amendment 1 of Annex 5 EPRO-FDHA, section 1.11

2.2.3 Actors / Transactions

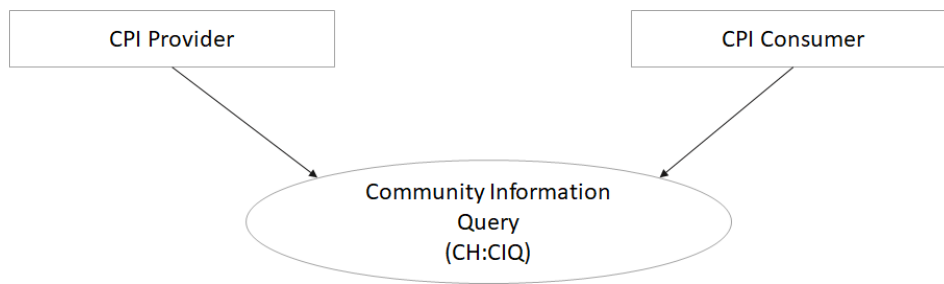


Figure 2: CH:CPI Actors for Community Information Query (CH:CIQ)

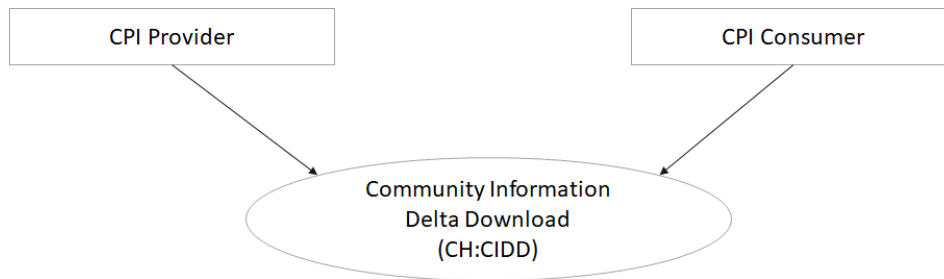


Figure 3: CH:CPI Actors for Community Information Delta Download (CH:CIDD)

Actor	Role
CH:CPI Consumer	This actor queries for information of the CH:CPI by either a query of the full content or a query for a delta download.
CH:CPI Provider	This actor provides information of the CH:CPI.

Table 1: Actor Roles of the CH:CPI profile

Actor	Transaction	Opt.	Reference
CH:CPI Consumer	Community Information Query [CH:CIQ]	R	3.1.4
	Community Information Delta Download [CH:CIDD]	O	3.1.6
CH:CPI Provider	Community Information Query [CH:CIQ]	R	3.1.5
	Community Information Delta Download [CH:CIDD]	R	3.1.7

Table 2: Actors and Transactions of the CH:CPI profile

2.2.4 Dependencies among Integration Profiles

Each CH:CPI Actor shall be grouped with the following actors: Secure Node or Secure Application (IHE ATNA) and with Time Client (IHE CT) including the corresponding Swiss National Extensions.

3 Volume 2 – Transactions

3.1 Community Portal Index (CH:CPI)

3.1.1 Scope

The CH:CPI transactions are used by the technical configuration component of a community to query the CH:CPI Provider in order to get information of the CH:CPI. This information allows a community to get the necessary information for communication with all certified communities within the Swiss EPR circle of trust. The CH:CPI contains names, endpoints and certificates that are used for setting up trustworthy communication links and other information such as contact data.

3.1.2 Referenced Standards

- a. Health informatics - Directory services for health care providers, subjects of care, and other entities (ISO/TS 21091)
- b. IETF LDAP v3 [RFC 2068, RFC 2251, RFC 2256, RFC 2985, RFC 2798, RFC 2985, 1045 RFC 5646, RFC 4512, RFC 4517]
- c. OASIS Directory Services Markup Language (DSML) v2.0:
<https://www.oasis-open.org/committees/dsml/docs/DSMLv2.xsd>
- d. W3C SOAP Version 1.2
<https://www.w3.org/TR/soap12/>

3.1.3 Interaction Diagram

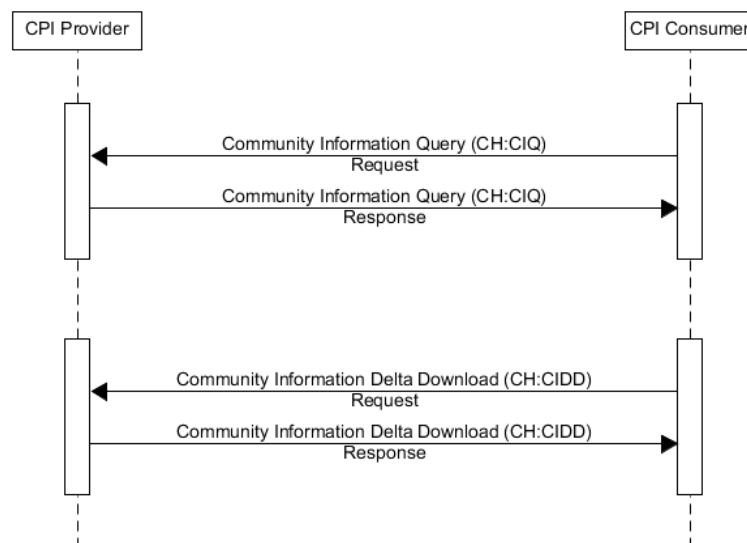


Figure 4: CH:CPI Interaction Diagram

3.1.4 Community Information Query (CH:CIQ) Request

3.1.4.1 Trigger Events

This request message is sent each time the CH:CPI Consumer needs a set of the currently available community information of the CH:CPI.

3.1.4.2 Message Semantics

The CH:CIQ Request contains a SOAP based DSMLv2 batchRequest message to express a query. See also Appendix A – Community Information Query schema (DSMLv2.xsd)

The SOAP operation namespace SHALL read: **ch:admin:bag:epr:CPI:2017**

The query operation for looking up entries in the CH:CPI SHALL be done through the use of searchRequest operation. The request SHALL be valid according to the DSML v2 schema. The CH:CPI Provider SHALL support all LDAP standard search filters. This transaction SHALL not limit any restriction on the search scope, time limit or list of attributes. The size limit SHALL be set to max. 1'000 result entries.

The following table indicates the object classes that can be queried.

Type	Element	LDAP objectClass
Community	Community	CHCommunity
Endpoint	XCA Initiating Gateway	CHXcalInitGw
Endpoint	XCA Responding Gateway	CHXcaRespGW
Endpoint	XCPD Initiating Gateway	CHXcpdInitGw
Endpoint	XCPD Responding Gateway	CHXcpdRespGw
Endpoint	CH:ADR Authorization Decision Consumer	CHAuDecCons
Endpoint	CH:ADR Authorization Decision Provider	CHAuDecProv
Endpoint	XUA X-Assertion Provider	CHAssertProv
Endpoint	CH:ATC Patient Audit Record Repository	CHAudRecRep
Endpoint	CH:ATC Patient Audit Consumer	CHPatAudCons
Endpoint	RMU Initiating Gateway	CHRmulInitGw
Endpoint	RMU Responding Gateway	CHRmuResGw
Endpoint	XCA-I Initiating Imaging Gateway	CHXcaiInitGw
Endpoint	XCA-I Responding Imaging Gateway	CHXcaiRespGW
Endpoint	MHD Document Source	CHMhdDocSrc
Endpoint	MHD Document Consumer	CHMhdDocCons
Endpoint	MHD Document Responder	CHMhdDocResp
Endpoint	IUA Authorization Server	CHAuthServer

Table 3: LDAP base objects

The following example requests the full CH:CPI information:

```
<soap:Envelope xmlns:soap=http://www.w3.org/2003/05/soap-envelope
  xmlns:a=http://www.w3.org/2005/08/addressing
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Header>
    <a:Action soap:mustUnderstand="1">
      urn:ch:admin:bag:epr:2017:CommunityQuery
    </a:Action>
    <a:To soap:mustUnderstand="1">YourServiceEndpointURL</a:To>
  </soap:Header>
  <soap:Body>
    <batchRequest xmlns="urn:oasis:names:tc:DSML:2:0:core">
      <searchRequest dn="DC=CPI,O=BAG,C=CH" scope="wholeSubtree"
        derefAliases="neverDerefAliases">
        <filter>
          <present name="objectClass"/>
        </filter>
      </searchRequest>
    </batchRequest>
  </soap:Body>
</soap:Envelope>
```

Figure 5: CH:CIQ Request for full CH:CPI information

3.1.4.3 Expected Actions

The CH:CPI Provider SHALL validate the authenticity of the CH:CPI Consumer by checking the X.509 certificate, as it is under no circumstances allowed to respond to a request from a source that is not part of the CH:CPI itself. This SHALL be done using the IHE ATNA Authenticate Node [ITI-19] transaction. Each node authentication failure (e.g. invalid X.509 certificate) SHALL trigger an IHE ATNA "Security Alert" Audit Event.

For legitimate requests, the CH:CPI Provider responds using the Community Information Query (CH:CIQ) Response message (see 3.1.5). For all other requests, the transaction SHALL fail (see 3.1.8).

3.1.5 Community Information Query (CH:CIQ) Response

3.1.5.1 Trigger Events

This response message is sent each time the CH:CPI Provider receives a CH:CIQ Request providing a valid X.509 certificate. The response either contains all entries that correspond to the searchRequest operation sent by the CH:CPI Consumer or in case of errors, the corresponding error message (SOAP fault or DSML ErrorResponse).

3.1.5.2 Message Semantics

3.1.5.2.1 SOAP fault

The CH:CIQ Response contains a SOAP fault in case of an invalid SOAP request.

The following example message shows a SOAP fault caused by an invalid SOAP header in the request:

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope">
  <s:Body>
    <s:Fault>
      <s:Code>
        <s:Value>s:Sender</s:Value>
      </s:Code>
      <s:Reason>
        <s:Text xml:lang="en-US">Unexpected internal server
error</s:Text>
      </s:Reason>
    </s:Fault>
  </s:Body>
</s:Envelope>
```

Figure 6: Sample CH:CIQ Response to an invalid SOAP header in the request

The CH:CIQ Response contains a SOAP fault in any case of an invalid SOAP body. I.e. when the DSML v2 XML schema is violated.

The following example message shows a SOAP fault caused by an invalid SOAP body in the request:

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope">
  <s:Body>
    <s:Fault>
      <s:Code>
        <s:Value>s:Sender</s:Value>
        <s:Subcode>
          <s:Value xmlns:a="urn:ch:admin:bag:epr:2017">
            a:XML_SCHEMA_VIOLATION
          </s:Value>
        </s:Subcode>
      </s:Code>
      <s:Reason>
        <s:Text xml:lang="en-US">[Explanation of the schema
violation]</s:Text>
      </s:Reason>
    </s:Fault>
  </s:Body>
</s:Envelope>
```

Figure 7: Sample CH:CIQ Response to an invalid SOAP body in the request

3.1.5.2.2 DSML ErrorResponse

The CH:CIQ Response contains a SOAP based DSMLv2 batchResponse message of errorResponse element type to send a response for an invalid searchRequest. See also Appendix A – Community Information Query schema (DSMLv2.xsd).

The following example message shows a DSML error response caused by an invalid DSML batchRequest:

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
  xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
      urn:ch:admin:bag:epr:2017:CommunityQueryResponse
    </a:Action>
  </s:Header>
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <batchResponse requestID=""
  xmlns="urn:oasis:names:tc:DSML:2:0:core">
      <errorResponse requestID="" type="malformedRequest"/>
    </batchResponse>
  </s:Body>
</s:Envelope>
```

Figure 8: Sample CH:CIQ Response to an invalid DSML batchRequest

3.1.5.2.3 DSML SearchResponse

The CH:CIQ Response contains a SOAP based DSMLv2 batchResponse message of searchResponse element type to send a response for a valid searchRequest. See also Appendix A – Community Information Query schema (DSMLv2.xsd).

The CH:CIQ Response SHALL contain the batchRequest and searchRequest requestID attributes of the corresponding CH:CIQ Request.

The attribute "shcStatus" of a community SHALL be respected. It contains either "active" or "inactive". Only community elements having "shcStatus=active" are to be considered as part of the Swiss EPR circle of trust. Their configuration data is up to date. Community elements having "shcStatus=inactive" are to be considered as outside of the Swiss EPR circle of trust. They are therefore no longer trustworthy and all communication from and to these communities is forbidden.

The relationships between communities and their endpoints can be looked up by querying the following attributes of CHCommunity object classes:

- a. shcXcalniGW: IHE XCA Initiating Gateways
- b. shcXcaRespGW: IHE XCA Responding Gateways
- c. shcXcpdIniGW: IHE XCPD Initiating Gateways
- d. shcXcpdResGW: IHE XCPD Responding Gateways
- e. shcAuDecProv: CH:ADR Authorization Decision Provider
- f. shcAuDecCons: CH:ADR Authorization Decision Consumer
- g. shcAsPrIsCrt: IHE XUA X-Assertion Provider
- h. shcAudRecRep: CH:ATC Patient Audit Record Repository
- i. shcPatAudCons: CH:ATC Patient Audit Consumer
- j. shcRmulnitGW: IHE RMU Initiating Gateway
- k. shcRmuResGW: IHE RMU Responding Gateway
- l. shcXcailniGW: IHE XCA-I Initiating Imaging Gateways
- m. shcXcaiRespGW: IHE XCA-I Responding Imaging Gateways
- n. shcMhdDocSrc: IHE MHD Document Source
- o. shcMhdDocCons: IHE MHD Document Consumer
- p. shcMhdDocResp: IHE MHD Document Responder
- q. shcAuthSrvIsCrt: IHE IUA Authorization Server

The following example message shows a DSML search response (fragment only):

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
      urn:ch:admin:bag:epr:2017:CommunityQueryResponse
    </a:Action>
  </s:Header>
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <batchResponse requestID=""
xmlns="urn:oasis:names:tc:DSML:2:0:core">
      <searchResponse requestID="test1">

        <searchResultEntry
dn="uid=CommunityA,OU=CHCommunity,DC=CPI,O=BAG,C=ch">
          <attr name="shcDisplayName">
            <value>Com-A</value>
          </attr>
          <attr name="shcIssuerName">
            <value>ComA</value>
          </attr>
          <attr name="objectClass">
            <value>top</value>
            <value>CHCommunity</value>
          </attr>
          <attr name="shcStatus">
            <value>Active</value>
          </attr>
          <!-- further attr elements omitted -->
        </searchResultEntry>
        <!-- further searchResultEntry elements omitted -->
      </searchResultEntry>
      <searchResultDone>
        <resultCode code="0"/>
      </searchResultDone>
    </searchResponse>
  </batchResponse>
</s:Body>
</s:Envelope>
```

Figure 9: Sample CH:CIQ Response (fragment only)

3.1.5.3 Expected Actions

The CH:CPI Consumer SHALL validate the authenticity of the CH:CPI Provider by checking the X.509 certificate, as it is under no circumstances allowed to process responses from a source that is not the CH:CPI Provider itself. This SHALL be done using the IHE ATNA Authenticate Node [ITI-19] transaction. Each node authentication failure (e.g. invalid X.509 certificate) SHALL trigger an IHE ATNA "Security Alert" Audit Event.

For legitimate responses, the CH:CPI Consumer processes the content of the response (see 4.1). For all other requests, the response SHALL be discarded (see 3.1.8).

3.1.6 Community Information Delta Download (CH:CIDD) Request

3.1.6.1 Trigger Events

This request message is sent each time the CH:CPI Consumer needs a set of modified community information entries in the CH:CPI. Therefore, this request message allows to specify a time interval. The CH:CPI Consumer expects therefore a response containing all entries on the CH:CPI that have been updated within this time span.

3.1.6.2 Message Semantics

The CH:CIDD Request contains a SOAP based downloadRequest message to express a delta download query. See also Appendix B – Community Information Delta Download schema (CIDD.xsd).

The SOAP operation namespace SHALL read: **ch:admin:bag:epr:CPI:2017**

The delta download query operation for looking up entries in the CH:CPI SHALL be done through the use of downloadRequest operation. The following table describes the request parameters.

Parameter name	Opt.	Description
fromDate	R	Lower boundary (including beginning) of the desired time span according to W3C dateTime datatype (YYYY-MM-DDThh:mm:ss.sTZD).
toDate	O	Upper boundary (including end) of the desired time span according to W3C dateTime datatype (YYYY-MM-DDThh:mm:ss.sTZD). If not provided, the current serverside timestamp is used (see Annex 2 EPRO-FDHA, chapter 2.9.30).
requestID	O	An optional requestID according to W3C string data type that allows the association of the request in the response.

Table 4: CH:CIDD request parameters

The following example requests all CH:CPI information updates in January 2018:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
  xmlns:a="http://www.w3.org/2005/08/addressing"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Header>
    <a:Action soap:mustUnderstand="1">
      urn:ch:admin:bag:epr:2017:CommunityDownload
    </a:Action>
    <a:To soap:mustUnderstand="1">YourServiceEndpointURL</a:To>
  </soap:Header>
  <soap:Body>
    <downloadRequest xmlns="urn:ch:admin:bag:epr:2017"
      fromDate="2018-01-01T00:00:00.000Z"
      toDate="2018-01-31T23:59:59.999Z" />
  </soap:Body>
</soap:Envelope>
```

Figure 10: CH:CIDD Request for information modified within a given time span

3.1.6.3 Expected Actions

The CH:CPI Provider SHALL validate the authenticity of the CH:CPI Consumer by checking the X.509 certificate, as it is under no circumstances allowed to respond to a request from a source that is not part of the CH:CPI itself. This SHALL be done using the IHE ATNA Authenticate Node [ITI-19] transaction. Each node authentication failure (e.g. invalid X.509 certificate) SHALL trigger an IHE ATNA "Security Alert" Audit Event.

For legitimate requests, the CH:CPI Provider responds using the Community Information Delta Download (CH:CIDD) Response message (see 3.1.7). For all other requests, the transaction SHALL fail (see 3.1.8).

3.1.7 Community Information Delta Download (CH:CIDD) Response

3.1.7.1 Trigger Events

This response message is sent each time the CH:CPI Provider receives a CH:CIDD Request providing a valid X.509 certificate. It is the response either containing all entries that correspond to the downloadRequest operation sent by the CH:CPI Consumer or in case of errors, the corresponding error message (SOAP fault or DSML ErrorResponse).

3.1.7.2 Message Semantics

3.1.7.2.1 SOAP fault

The CH:CIDD Response contains a SOAP fault in case of an invalid SOAP or download request.

For an example to an invalid SOAP request, see Figure 6. The following example message shows a SOAP fault caused by a missing downloadRequest:

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
            xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
      http://www.w3.org/2005/08/addressing/soap/fault
    </a:Action>
  </s:Header>
  <s:Body>
    <s:Fault>
      <s:Code>
        <s:Value>s:Sender</s:Value>
      </s:Code>
      <s:Reason>
        <s:Text xml:lang="en-US">
          The delta download request is not specified.
        </s:Text>
      </s:Reason>
    </s:Fault>
  </s:Body>
</s:Envelope>
```

Figure 11: Sample CH:CIDD Response to an invalid SOAP header in the request

The CH:CIDD Response contains a SOAP fault in any case of an invalid SOAP body. I.e. when the DSML v2 XML schema is violated.

For an example message containing a SOAP fault caused by an invalid SOAP body in the request, see Figure 7.

3.1.7.2.2 DSML ErrorResponse

The CH:CIDD Response SHALL NOT use SOAP based DSMLv2 batchResponse message containing the errorResponse element. SOAP faults SHALL be used, instead (see 3.1.7.2.1).

3.1.7.2.3 CH:CIDD DownloadResponse

The CH:CIDD Response contains a SOAP based downloadResponse message of DSMLv2 batchRequest element type to send a response for a valid downloadRequest.

See also:

- a. DownloadResponse:
Appendix B – Community Information Delta Download schema (CIDD.xsd).
- b. DSMLv2 batchRequest:
Appendix A – Community Information Query schema (DSMLv2.xsd).

The CH:CIDD Response SHALL contain the requestID to associate it to the corresponding CH:CIDD Request.

The CH:CPI Provider SHALL ensure that the following main elements are mutually exclusive within the CH:CIDD Response.

Element name	Opt.	Description
addRequest	O	Indicates a new CH:CPI information entry and delivers full CH:CPI information for this element. It contains the same elements as described for CH:CIQ Response. See 0 and 4.1.3.
modifyRequest	O	Indicates an updated CH:CPI information entry and delivers the changed information, only. This allows the CH:CPI Consumer to update its information base. It contains a <i>modification</i> element having the <i>name</i> attribute set with the name of the changed CH:CPI information (e.g. "shcGatewayCert") and the <i>operation</i> attribute is set to "replace". The modification element contains two child elements as instances of the <i>value</i> element: - The first instance contains the value before the update - The second instance contains the value after the update
modDNRequest	O	Indicates the change of the distinguished name of a CH:CPI entry.
delRequest	O	Indicates the removal of a previous CH:CPI information entry. It does not contain any further elements. The deleted entry is identified by the dn attribute.

Table 5: CH:CIDD response main elements

The following example message shows a CH:CIDD Response (fragment only):

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
      urn:ch:admin:bag:epr:2017:CommunityDownloadResponse
    </a:Action>
  </s:Header>
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <downloadResponse requestID="test2"
xmlns="urn:ch:admin:bag:epr:2017">
      <batchRequest onError="resume"
xmlns="urn:oasis:names:tc:DSML:2:0:core">
        <addRequest requestID="2017-12-11T11:55:31.7643345Z"

dn="uid=newEntry,OU=CHCommunity,DC=CPI,O=BAG,C=ch">
          <!-- details ommited -->
        </addRequest>
      </batchRequest>
      <batchRequest onError="resume"
xmlns="urn:oasis:names:tc:DSML:2:0:core">
        <addRequest requestID="2017-12-12T08:09:52.7154691Z"
          dn="uid=newEntry:XcaInitiatingGateway,
            OU=CHEndpoint,DC=CPI,O=BAG,C=ch">
          <!-- details ommited -->
        </addRequest>
        <modifyRequest requestID="2017-12-12T08:09:52.7464564Z"

dn="uid=existCmty1,OU=CHCommunity,DC=CPI,O=BAG,C=ch">
          <!-- details ommited -->
        </modifyRequest>
      </batchRequest>
      <batchRequest onError="resume"
xmlns="urn:oasis:names:tc:DSML:2:0:core">
        <modifyRequest requestID="2017-12-12T08:10:19.2713348Z"

dn="uid=existCmty2,OU=CHCommunity,DC=CPI,O=BAG,C=ch">
          <!-- details ommited -->
        </modifyRequest>
      </batchRequest>
      <batchRequest onError="resume"
xmlns="urn:oasis:names:tc:DSML:2:0:core">
        <delRequest requestID="2017-12-12T13:35:53.2923242Z"
          dn="uid=existCmty3:XcaInitiatingGateway,
            OU=CHEndpoint,DC=CPI,O=BAG,C=ch"/>
      </batchRequest>
    </downloadResponse>
  </s:Body>
</s:Envelope>
```

Figure 12: Sample CH:CIDD Response (fragment only)

3.1.7.3 Expected Actions

The expected actions for CH:CIDD Response are identical to those of CH:CIQ. See 3.1.5.3.

3.1.8 Security Considerations

The IHE HPD Integration profile is taken as basic for the CH:CPI profile. The CH:CPI profile requires all actors to be grouped with Secure Node or Secure Application implementing the "STX: TLS 1.2 floor using BCP195 Option" defined in the IHE ITI TF-2³, chapter 3.19.6.2.3. The communication requires X.509 certificates from the "Swiss Government Root CA II". Beyond this, the CH:CPI profile requires all security considerations from the IHE HPD Integration profile.

3.1.9 Audit Record Considerations

The Actors involved SHALL record audit events according to the following audit messages.

3.1.9.1 CH:CPI Provider Audit Messages

3.1.9.1.1 CH:CPI Provider Audit Message for CH:CIQ

Event	Field Name	Value constraints
Event	EventID	EV(000001, BAG, "CH:CIQ")
	EventActionCode	"R" (Read)
	EventDateTime	UTC-Time
	EventOutcomeIndicator	Success 0, Failure 4
	EventTypeCode	EV("CH:CIQ", "CH:EPR Transactions", "Community Information Query")
Source	UserID	Community with prefix (shclssuerName)
	UsersRequestor	"true"
	RoleIDCode	EV(110153, DCM, "Source Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Destination	UserID	SOAP endpoint URI
	AlternativeUserID	The process ID as used within the local operating system in the local system logs
	UsersRequestor	"false"
	RoleIDCode	EV(110152, DCM, "Destination Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Audit Source	AuditSourceID	CPI
	AuditEnterpriseSiteID	EPD-AD.BAG.admin.ch
	AuditSourceTypeCode	"4" for Application server process tier in a multi-tier system
Community	ParticipantObjectTypeCode	"2" system object
	ParticipantObjectTypeCodeRole	"24" query
	ParticipantObjectDataLifeCycle	"6" access use
	ParticipantObjectIDTypeCode	EV("CH:CIQ", "CH:EPR Transactions", "Community Information Query")
	ParticipantObjectID	The requestID of the searchRequest
	ParticipantObjectDetail	All CI-attributes of the searchRequest

Table 6: CH:CPI Provider Audit Message for CH:CIQ

³ IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

3.1.9.1.2 CH:CPI Provider Audit Message for CH:CIDD

Event	Field Name	Value constraints
Event	EventID	EV(000006, BAG, "CH:CIDD")
	EventActionCode	"R" (Read)
	EventDateTime	UTC-Time
	EventOutcomeIndicator	Success 0, Failure 4
	EventTypeCode	EV("CH:CIDD", "CH:EPR Transactions", "Community Information Delta Download")
Source	UserID	Community with prefix (shclssuerName)
	UserIsRequestor	"true"
	RoleIDCode	EV(110153, DCM, "Source Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Destination	UserID	SOAP endpoint URI
	AlternativeUserID	The process ID as used within the local operating system in the local system logs
	UserIsRequestor	"false"
	RoleIDCode	EV(110152, DCM, "Destination Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Audit Source	AuditSourceID	CPI
	AuditEnterpriseSiteID	EPD-AD.BAG.admin.ch
	AuditSourceTypeCode	"4" for Application server process tier in a multi-tier system
Community	ParticipantObjectTypeCode	"2" system object
	ParticipantObjectTypeCodeRole	"24" query
	ParticipantObjectDataLifeCycle	"6" access or use
	ParticipantObjectIDTypeCode	EV("CH:CIDD", "CH:EPR Transactions", "Community Information Delta Download")
	ParticipantObjectID	The requestID of the downloadRequest
	ParticipantObjectDetail	The CH:CIDD downloadRequest parameters: – fromDate – toDate – requestID

Table 7: CH:CPI Provider Audit Message for CH:CIDD

3.1.9.2 CH:CPI Consumer Audit Messages

3.1.9.2.1 CH:CPI Consumer Audit Message for CIQ

Event	Field Name	Value constraints
Event	EventID	EV(000001, BAG, "CH:CIQ")
	EventActionCode	"R" (Read)
	EventDateTime	UTC-Time
	EventOutcomeIndicator	Success 0, Failure 4
	EventTypeCode	EV("CH:CIQ", "CH:EPR Transactions", "Community Information Query")
Source	UserID	The process ID as used within the local operating system in the local system logs.
	UsersRequestor	"true"
	RoleIDCode	EV(110153, DCM, "Source Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Destination	UserID	SOAP endpoint URI
	AlternativeUserID	Not specialized
	UsersRequestor	"false"
	RoleIDCode	EV(110152, DCM, "Destination Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Audit Source	AuditSourceID	Not specialized
	AuditEnterpriseSiteID	Not specialized
	AuditSourceTypeCode	Not specialized
Consumer	ParticipantObjectTypeCode	"2" system object
	ParticipantObjectTypeCodeRole	"24" query
	ParticipantObjectDataLifeCycle	"6" access use
	ParticipantObjectIDTypeCode	EV("CH:CIQ", "CH:EPR Transactions", "Community Information Query")
	ParticipantObjectID	The requestID of the searchRequest
	ParticipantObjectDetail	All CI-attributes of the searchRequest

Table 8: CH:CPI Consumer Audit Message for CH:CIQ

3.1.9.2.2 CH:CPI Consumer Audit Message for CIDD

Event	Field Name	Value constraints
Event	EventID	EV(000006, BAG, "CH:CIDD")
	EventActionCode	"R" (Read)
	EventDateTime	UTC-Time
	EventOutcomeIndicator	Success 0, Failure 4
	EventTypeCode	EV("CH:CIDD", "CH:EPR Transactions", "Community Information Delta Download")
Source	UserID	The process ID as used within the local operating system in the local system logs.
	UsersRequestor	"true"
	RoleIDCode	EV(110153, DCM, "Source Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Destination	UserID	SOAP endpoint URI
	AlternativeUserID	Not specialized
	UsersRequestor	"false"
	RoleIDCode	EV(110152, DCM, "Destination Role ID")
	NetworkAccessPointTypeCode	"2" for IP address
	NetworkAccessPointID	The IP address
Audit Source	AuditSourceID	Not specialized
	AuditEnterpriseSiteID	Not specialized
	AuditSourceTypeCode	Not specialized
Consumer	ParticipantObjectTypeCode	"2" system object
	ParticipantObjectTypeCodeRole	"24" query
	ParticipantObjectDataLifeCycle	"6" access or use
	ParticipantObjectIDTypeCode	EV("CH:CIDD", "CH:EPR Transactions", "Community Information Delta Download")
	ParticipantObjectID	The requestID of the downloadRequest
	ParticipantObjectDetail	The CH:CIDD downloadRequest parameters: fromDate toDate requestID

Table 9: CH:CPI Consumer Audit Message for CH:CIDD

4 Volume 3 – Content Profile

4.1 Community Information Format

4.1.1 Scope

The CH:CPI requires to express different attributes for a community and its endpoints. As the concept described in Volume 2 follows the IHE HPD profile⁴, CH:CPI is also based on LDAP. This content profile therefore describes the detailed usage of the LDAP structure and attributes for the usage in CH:CPI.

This content profile does not explain the spirit of LDAP but only focuses on the detailed field description.

DSML v2 expresses LDAP requests and responses as XML document fragments. This specification therefore only specifies the usage of DSML v2.

4.1.2 Referenced Standards

OASIS Directory Services Markup Language (DSML) v2.0:

<https://www.oasis-open.org/committees/dsml/docs/DSMLv2.xsd>

4.1.3 Detailed CH:CPI Format definitions

The following illustration shows the LDAP structure for CH:CPI. It is not complete and therefore intended as an overview, only. See sub sections for the detailed attribute lists.

⁴ See Amendment 1 of Annex 5 EPRO-FDHA

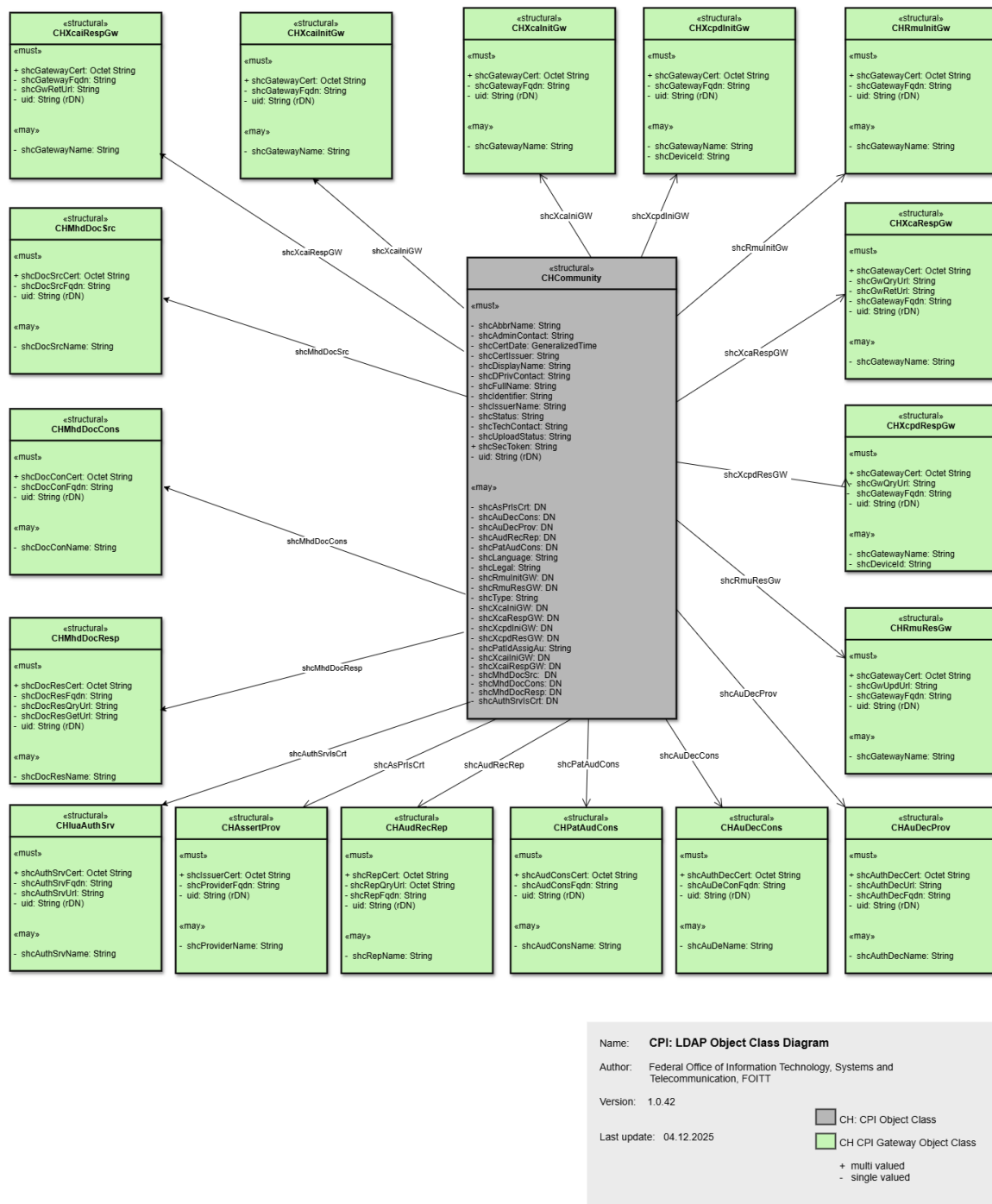


Figure 13: CH:CPI LDAP Schema

4.1.3.1 Conventions

Conventions for the tables in the following sections:

Optionality column (Opt:)

- a. O=optional
- b. R=required

Cardinality column (Card):

- c. S=Single-valued
- d. M=Multi-valued

4.1.3.2 Community

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Community rDN
shcFullName	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.1	Community full name
shcAbbrName	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.2	Community abbreviated name
shcDisplayName	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.3	Community display name
shcLegal	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.4	Legal form of the community (see 4.1.4.1)
shcIssuerName	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.15	Community issuer name
shcIdentifier	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.6	Community OID
shcPatIdAssigAu	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.39	Patient ID assigning authority OID
shcAdminContact	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.5	Administrative contact information of the community
shcTechContact	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.7	Technical contact information of the community
shcDPrivContact	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.8	Data privacy contact information of the community
shcCertDate	R	S	GeneralizedTime	2.16.756.5.30.1.127.3.10.4.10	Latest date of community certification according to Annex 2 EPRO-FDHA
shcType	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.14	Community type
shcCertIssuer	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.9	Latest certification issuer of the community
shcLanguage	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.11	Community main language
shcStatus	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.12	Community status. Only active communities build the Swiss EPR circle of trust
shcUploadStatus	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.13	Upload status of a community
shcSecToken	R	M	DirectoryString	2.16.756.5.30.1.127.3.10.4.17	Security tokens of a community to access the EPR central services
shcXcalniGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.18	DN of the community XCA Initiating Gateway element
shcXcaRespGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.20	DN of the community XCA Responding Gateway element
shcXcpdIniGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.22	DN of the community XCPD Initiating Gateway element
shcXcpdResGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.24	DN of the community XCPD Responding Gateway element
shcAuDecProv	O	S	DN	2.16.756.5.30.1.127.3.10.4.26	DN of the community CH:ADR Authorization Decision Provider element

Attribute name	Opt.	Card.	Data type	OID	Description
shcAuDecCons	O	S	DN	2.16.756.5.30.1.127.3.10.4.28	DN of the community CH:ADR Authorization Decision Consumer element
shcAsPrIsCrt	O	S	DN	2.16.756.5.30.1.127.3.10.4.30	DN of the community XUA X-Assertion Provider Issuer Certificate element
shcAudRecRep	O	S	DN	2.16.756.5.30.1.127.3.10.4.56	DN of the community CH:ATC Patient Audit Record Repository element
shcPatAudCons	O	S	DN	2.16.756.5.30.1.127.3.10.4.65	DN of the community CH:ATC Patient Audit Consumer element
shcRmuInitGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.58	DN of the community RMU Initiating Gateway element
shcRmuResGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.60	DN of the community RMU Responding Gateway element
shcXcaiIniGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.69	DN of the community XCA-I Initiating Imaging Gateway element
shcXcaiRespGW	O	S	DN	2.16.756.5.30.1.127.3.10.4.70	DN of the community XCA-I Responding Imaging Gateway element
shcMhdDocSrc	O	S	DN	2.16.756.5.30.1.127.3.10.4.71	DN of the community MHD Document Source element
shcMhdDocCons	O	S	DN	2.16.756.5.30.1.127.3.10.4.72	DN of the community MHD Document Consumer element
shcMhdDocResp	O	S	DN	2.16.756.5.30.1.127.3.10.4.73	DN of the community MHD Document Responder element
shcAuthSrvIsCrt	O	S	DN	2.16.756.5.30.1.127.3.10.4.74	DN of the community IUA Authorization Server Issuer Certificate element

Table 10: CH:CPI content profile definitions for Communities

4.1.3.3 Endpoints

The following definition is valid for all subsequent endpoint types:

- a. Super Class="top"
- b. RDN=<uid>

4.1.3.3.1 XCA Initiating Gateway

OID: 2.16.756.5.30.1.127.3.10.4.32

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + XcaInitiatingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 11: CH:CPI content profile definitions for XCA Initiating Gateways

4.1.3.3.2 XCA Responding Gateway

OID: 2.16.756.5.30.1.127.3.10.4.33

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + XcaRespondingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGwQryUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.43	Query URL
shcGwRetUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.44	Retrieve URL
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 12: CH:CPI content profile definitions for XCA Responding Gateways

4.1.3.3.3 XCPD Initiating Gateway

OID: 2.16.756.5.30.1.127.3.10.4.36

Attribute name	Opt.	Card.	Data type	OID	Description
Uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" XcpdInitiatingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcDeviceId	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.40	Device Id OID
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 13: CH:CPI content profile definitions for XCPD Initiating Gateways

4.1.3.3.4 XCPD Responding Gateway

OID: 2.16.756.5.30.1.127.3.10.4.37

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" XcpdRespondingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGwQryUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.43	Query URL
shcDeviceId	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.40	Device Id OID
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 14: CH:CPI content profile definitions for XCPD Responding Gateways

4.1.3.3.5 CH:ADR Authorization Decision Provider

OID: 2.16.756.5.30.1.127.3.10.4.34

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + AuthorizationDecisionProviderGateway
shcAuthDecName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.49	CH:ADR Provider name
shcAuthDecUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.50	CH:ADR Provider URL
shcAuthDecFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.75	CH:ADR Provider FQDN
shcAuthDecCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.51	List of certificates identifying this endpoint

Table 15: CH:CPI content profile definitions for CH:ADR Authorization Decision Providers

4.1.3.3.6 ProvidersCH:ADR Authorization Decision Consumer

OID: 2.16.756.5.30.1.127.3.10.4.38

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + AuthorizationDecisionConsumerGateway
shcAuDeConName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.76	CH:ADR Consumer name
shcAuDeConFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.77	CH:ADR Consumer FQDN
shcAuthDecCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.51	List of certificates identifying this endpoint

Table 16: CH:CPI content profile definitions for CH:ADR Authorization Decision Consumers

4.1.3.3.7 XUA X-Assertion Provider

OID: 2.16.756.5.30.1.127.3.10.4.35

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + AssertionProviderIssuerCertificate
shcProviderName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.48	XUA X-Assertion Provider name
shcProviderFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.78	XUA X-Assertion Provider FQDN
shcIssuerCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.46	List of certificates used by this actor

Table 17: CH:CPI content profile definitions for XUA X-Assertion Providers

4.1.3.3.8 CH:ATC Patient Audit Record Repository

OID: 2.16.756.5.30.1.127.3.10.4.52

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + AtcPatientAuditRecordRepository
shcRepName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.53	Repository name
shcRepQryUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.54	Repository URL
shcRepFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.79	Repository FQDN
shcRepCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.55	List of certificates identifying this endpoint

Table 18: CH:CPI content profile definitions for CH:ATC Patient Audit Record Repositories

4.1.3.3.9 CH:ATC Patient Audit Consumer

OID: 2.16.756.5.30.1.127.3.10.4.66

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + AtcPatientAuditConsumer + optional suffix where "optional suffix" is "-" + <number>.

Attribute name	Opt.	Card.	Data type	OID	Description
shcAudConsName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.67	CH:ATC Patient Audit Consumer name
shcAudConsFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.80	CH:ATC Patient Audit Consumer FQDN
shcAudConsCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.68	List of certificates identifying this endpoint

Table 19: CH:CPI content profile definitions for CH:ATC Patient Audit Consumers

4.1.3.3.10 RMU Initiating Gateway

OID: 2.16.756.5.30.1.127.3.10.4.62

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" RmuInitiatingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 20: CH:CPI content profile definitions for RMU Initiating Gateways

4.1.3.3.11 RMU Responding Gateway

OID: 2.16.756.5.30.1.127.3.10.4.63

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + RmuRespondingGateway
shcGateway-Name	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGwUpdUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.64	Update URL
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 21: CH:CPI content profile definitions for RMU Responding Gateways

4.1.3.3.12 XCA-I Initiating Imaging Gateway

OID: 2.16.756.5.30.1.127.3.10.4.81

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + XcaiInitiatingImagingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 22: CH:CPI content profile definitions for XCA-I Initiating Imaging Gateways

4.1.3.3.13 XCA-I Responding Imaging Gateway

OID: 2.16.756.5.30.1.127.3.10.4.82

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + XcaiRespondingImagingGateway
shcGatewayName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.41	Gateway name
shcGwRetUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.44	Gateway Retrieve URL
shcGatewayFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.42	Gateway FQDN
shcGatewayCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.47	List of certificates identifying this endpoint

Table 23: CH:CPI content profile definitions for XCA-I Responding Imaging Gateways

4.1.3.3.14 MHD Document Source

OID: 2.16.756.5.30.1.127.3.10.4.83

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shclssuerName + ":" + MhdDocumentSource + optional suffix where "optional suffix" is "-" + <number>
shcDocSrcName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.87	MHD Document Source name
shcDocSrcFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.88	MHD Document Source FQDN
shcDocSrcCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.89	List of certificates identifying this endpoint

Table 24: CH:CPI content profile definitions for MHD Document Source

4.1.3.3.15 MHD Document Consumer

OID: 2.16.756.5.30.1.127.3.10.4.84

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shclssuerName + ":" + MhdDocumentConsumer + optional suffix where "optional suffix" is "-" + <number>
shcDocConName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.90	MHD Document Consumer name
shcDocConFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.91	MHD Document Consumer FQDN
shcDocConCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.92	List of certificates identifying this endpoint

Table 25: CH:CPI content profile definitions for MHD Document Consumers

4.1.3.3.16 MHD Document Responder

OID: 2.16.756.5.30.1.127.3.10.4.85

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + MhdDocumentResponder
shcDocResName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.93	MHD Document Responder name
shcDocResQryUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.94	MHD Document Responder FHIR Base URL
shcDocResGetUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.95	MHD Document Responder HTTP GET URL for Retrieve Document [ITI-68]
shcDocConFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.96	MHD Document Responder FQDN
shcDocResCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.97	List of certificates identifying this endpoint

Table 26: CH:CPI content profile definitions for MHD Document Responders

4.1.3.3.17 IUA Authorization Server

OID: 2.16.756.5.30.1.127.3.10.4.86

Attribute name	Opt.	Card.	Data type	OID	Description
uid	R	S	DirectoryString	0.9.2342.19200300.100.1.1	Endpoint rDN. Concatenation rule: CHCommunity.shcIssuerName + ":" + AuthorizationServerIssuerCertificate
shcAuthSrvName	O	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.98	IUA Authorization Server name
shcAuthSrvUrl	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.99	IUA Authorization Server HTTP GET URL for Get Authorization Server Metadata [ITI-103]
shcAuthSrvFqdn	R	S	DirectoryString	2.16.756.5.30.1.127.3.10.4.100	IUA Authorization Server FQDN
shcAuthSrvCert	R	M	OctetString	2.16.756.5.30.1.127.3.10.4.101	List of certificates used by this actor

Table 27: CH:CPI content profile definitions for IUA Authorization Servers

4.1.4 Value sets

The value sets mentioned in the following subsections are static value sets. All entries of the following value sets are case sensitive.

4.1.4.1 shcLegal

This value set contains the final list of allowed entries describing the legal form of the community.

Code	Description	Group
CompanyLimitedByShares	ge: Aktiengesellschaft / AG fr: société anonyme / SA	Capital companies ge: Kapitalgesellschaften fr: sociétés de capitaux)
PartnershipLimitedByShares	ge: Kommanditaktiengesellschaft / Kommandit-AG fr: société en commandite par actions	
LimitedLiabilityCompany	ge: Gesellschaft mit beschränkter Haftung / GmbH fr: société à responsabilité limitée / Sarl.	
Cooperative	ge: Genossenschaft fr: société cooperative	Other legal forms
Foundation	ge: Stiftung fr: fondation	
Association	ge: Verein fr: Société / association	

Table 28: Value set for shcLegal

4.1.4.2 shcType

This value set contains the final list of allowed entries describing the type of a community.

Code	Description
ReferenceCommunity	ge: Stammgemeinschaft fr: communauté de référence
Community	ge: Gemeinschaft fr: communauté

Table 29: Value set for shcType

4.1.4.3 shcLanguage

This value set contains the final list of allowed entries describing the main language of a community.

Code	Description
de	German
fr	French
it	Italian

Table 30: Value set for shcLanguage

4.1.4.4 shcStatus

This value set contains the final list of allowed entries describing the community status.

Code	Description
Active	Community is part of the Swiss EPR circle of trust.
Inactive	Community is considered as untrusted.

Table 31: Value set for shcStatus

5 Appendix

5.1 Appendix A – Community Information Query schema (DSMLv2.xsd)

The Community Information Query is based on OASIS Directory Services Markup Language (DSML) v2.0. See section "3.1.2 Referenced Standards".

5.2 Appendix B – Community Information Delta Download schema (CIDD.xsd)

This schema makes use of the OASIS Directory Services Markup Language (DSML) v2.0 schema and defines the following additional elements:

- a. downloadRequest (containing the parameters fromDate, toDate, requestID)
- b. downloadResponse (refers to the DSML batchRequest element).

5.3 Appendix C – CH:CPI Webservice (WSDL)

The CH:CPI Webservice defines the following operations:

- a. CommunityQueryRequest for CH:CIQ
- b. CommunityDownloadRequest for CH:CIDD

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